INTERACTIVE PEDAGOGICAL TOOL VISUALIZATION FOR QUANTUM GATES

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Abstract – Learning quantum computing is challenging because quantum gates and state vectors are highly abstract and difficult to visualize. Most students encounter static diagrams or complex mathematical representations before understanding how a qubit state evolves. Although Qiskit provides powerful simulation capabilities, it lacks a simple and classroom-friendly visualization interface. This project presents an interactive pedagogical tool built using Qiskit, designed to improve conceptual understanding through visual intuition. The system uses Qiskit Aer for quantum simulation and a Tkinter-based graphical interface for gate selection and qubit reset. The qubit's evolution is visualized in real time using a Bloch sphere representation, while measurement outcomes are displayed through probability histograms. Learners can apply quantum gates step by step and immediately observe their effects, helping connect geometric interpretation with measurable results. Experimental testing demonstrates accurate single-qubit behavior and responsive visual feedback, making the tool effective for teaching and learning introductory quantum computing concepts.

Keywords—Quantum Computing, Qiskit, Quantum Gates, Bloch Sphere, State Vector Visualization, Quantum Simulation, Interactive Learning Tool